

The empirical recurrence for OEIS A300678, recovered from its tail

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Abstract

OEIS A300678 records “Empirical recurrence of order 82” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 98$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 85$. Its last coefficient is $c_{82} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-82 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300678 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 2, 3 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 10, 11, 28, 80, 245, 729, 2604, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Mar 11 2018 08:39:25, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 98$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 98$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 82: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 82.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 196$ exact terms from $n = 3$ on, it returns order 82 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	0	c_{22}	23217	c_{43}	622590	c_{64}	−1001
c_2	8	c_{23}	1390	c_{44}	−618010	c_{65}	1140
c_3	16	c_{24}	−37503	c_{45}	−951889	c_{66}	851
c_4	8	c_{25}	−18637	c_{46}	−428	c_{67}	3291
c_5	−29	c_{26}	−63783	c_{47}	−334501	c_{68}	126
c_6	−76	c_{27}	−1155	c_{48}	109594	c_{69}	−600
c_7	−116	c_{28}	41408	c_{49}	612230	c_{70}	1066
c_8	−434	c_{29}	155999	c_{50}	26180	c_{71}	−616
c_9	−733	c_{30}	350355	c_{51}	146467	c_{72}	−607
c_{10}	429	c_{31}	337073	c_{52}	28628	c_{73}	269
c_{11}	1672	c_{32}	397326	c_{53}	−288158	c_{74}	95
c_{12}	2516	c_{33}	−201766	c_{54}	−9489	c_{75}	−92
c_{13}	5148	c_{34}	−381522	c_{55}	−49528	c_{76}	26
c_{14}	4217	c_{35}	−564806	c_{56}	−80515	c_{77}	41
c_{15}	−38	c_{36}	−1040441	c_{57}	94214	c_{78}	−31
c_{16}	−1743	c_{37}	−365995	c_{58}	23228	c_{79}	−10
c_{17}	−10633	c_{38}	−67562	c_{59}	7696	c_{80}	10
c_{18}	−21033	c_{39}	5019	c_{60}	21788	c_{81}	1
c_{19}	−15346	c_{40}	1137816	c_{61}	−6163	c_{82}	−1
c_{20}	−10607	c_{41}	986158	c_{62}	−15053		
c_{21}	−1137	c_{42}	−588	c_{63}	−9122		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A300678 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{82} - c_1 x^{82-1} - \dots - c_{82}$. Its constant term is $-c_{82} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 82 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 82, so $\deg q = 82$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 82, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 82, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-82 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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